Apollonian Circles

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1 Second Definition of Circle

1.1 Introduction

Since the sixth century B.C.E., Greek mathematicians have formalized and explored the study of geometry. It is well known that Euclid formally defined circles as the set of points equidistant from a given point at a given distance. However, people often overlook the second definition of circle by another excellent geometrician, Apollonius, with the definition being "the set of points that have a specific ratio of distance to two fixed points". The circle defined with Apollonius' definition is called "Apollonius Circle." A synthetic proof of that definition goes like the following:

1.2 Proof

Here, we will prove the statement that "the set of points with a specific ratio of distance to two fixed points is a circle." However, before we prove this statement, we will go over a few definitions and theorems.

1.2.1 Definitions

Definition 1.2.1 (Locus): A locus is a set of all points that satisfy a specific geometric rule, which often is a shape, curve, or line. A well-known example of a locus is a circle, which consists of all points equidistant from its center.

Definition 1.2.2 (Angle Bisector): An angle bisector is a line that bisects an angle into two congruent angles.

Definition 1.2.3 (Exterior Angle): An exterior angle is an angle formed by extending one of the lines that enclose the original angle.

Definition 1.2.4 (Exterior Angle Bisector): An exterior angle bisector is the line that bisects an exterior angle.

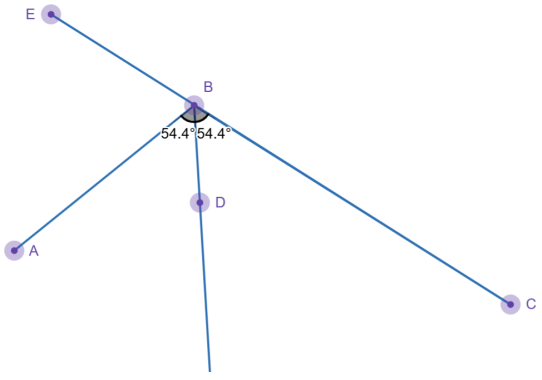


Figure 1: Angle ABE is the exterior angle of angle ABC, BD is the angle bisector of angle ABC

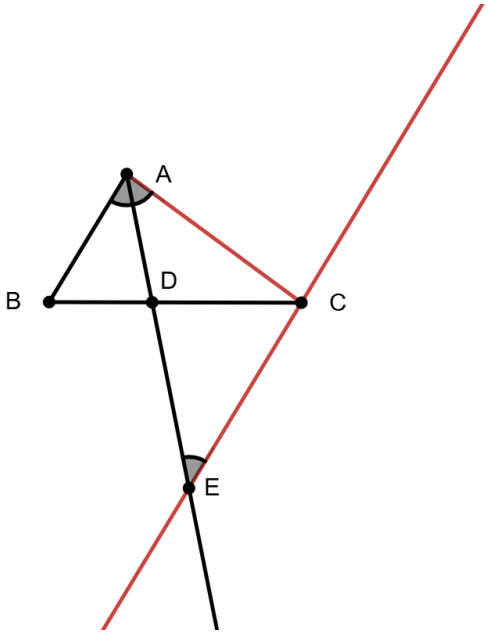


Figure 2: Angle Bisector

1.2.2 Theorems

Angle Bisector Theorem: The angle bisector theorem states that in $\triangle ABC$, if the angle bisector of $\angle BAC$ intersects BC at point D, then $\frac{AB}{BD} = \frac{AC}{CD}$

There are several ways to prove this theorem; here, we will do an entirely synthetic proof that goes as follows:

We will first construct a line intersecting point C that is parallel to line AB. We then extend AD until it intersects the line through point E that we constructed. $\angle BAD \cong \angle AEC$ because of $AB \parallel CE$, which means that $AC = CE$, since $\angle DAC = \angle DEC$. It is clear that $\triangle DAB \sim \triangle DEC$, therefore it is true that $\frac{AB}{BD} = \frac{CE}{DC}$. Substituting $AC = CE$ yields the desired result.

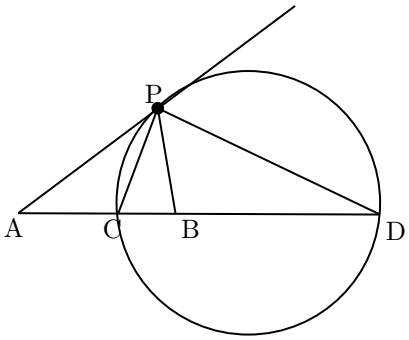
The Exterior Angle Bisector Theorem: The exterior angle bisector of a triangle divides the opposite side externally in the ratio of the adjacent sides. As the proof is similar to the proof of the angle bisector theorem, the proof will be left as an exercise for the reader.

1.2.3 Proof of the Statement

Let the two fixed points be point A and point B, and point P a point satisfying $\frac{PA}{PB} = k$, with k as a positive constant that is not 1

We first construct the angle bisector of the $\angle APB$ such that it intersects the line AB at point C. By the angle bisector theorem, $\frac{PA}{PB} = \frac{CA}{CB}$, which means that point C is a fixed point in the set. Then we will construct the exterior angle bisector of the $\angle APB$, which intersects the line AB at the point D. By the exterior angle bisector theorem, $\frac{PA}{PB} = \frac{AD}{BD}$, which means that point D is another point within the set. By the definitions of an angle bisector and an exterior angle bisector, $\angle CPD$ is always a right angle. By Thales' Theorem, the point P can be any point that lies on the circle with diameter CD.

The converse of the statement that we have proven: "all points on the circle satisfy the ratio $\frac{PA}{PB} = k$ for two fixed points A and B" is left as an exercise for the reader. (Hint: Angle Bisector Theorem)



1.2.4 Degenerate Case

We know that the definition of the locus is all points that satisfy $\frac{PA}{PB} = k$, with k as a constant. However, a degenerate circle will be constructed if $k = 1$, which is pretty intuitive. Since we want $PA = PB$, that is, the point P is on the perpendicular bisector of segment AB.

2 Applications

2.1 But why?

You may be wondering why we need a definition like this or how this is useful. This property of a circle allows us to solve the Apollonius Pursuit problem, which concerns minimizing the time of the chase.

The problem goes as follows: Given a ship moving in a constant direction and speed v, in what direction should the chase ship with speed V go, to intercept the other boat with minimal time spent?

The solution to this problem is also pretty simple. We want to find all points reachable by both ships simultaneously. Every t amount of time passes, the chasing ship will displace by Vt units of distance, while the evading ship will displace by vt units of distance. That means that the ratio of their displacement is a constant $\frac{V}{v}$. The set of locations that the two ships can simultaneously reach is a circle centered on the evading ship's location. Therefore, when the direction of the evading ship is given, the chasing boat has to move towards the area where the circle and the evading ship's direction intersect.

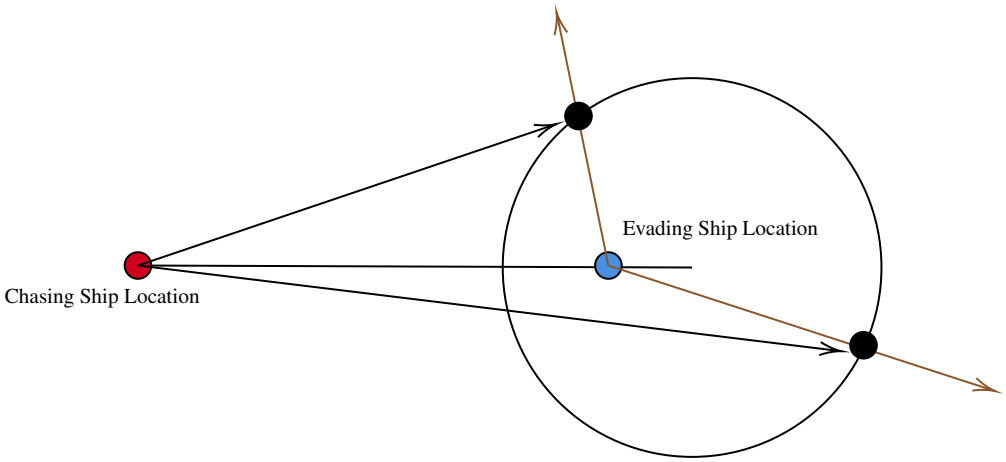


Figure 3 (The brown arrow is the direction the evading ship is heading, and the intersection is where the chasing ship should head.)

2.2 Exercise

Here is a problem that applies the Apollonius circle to real-life issues.

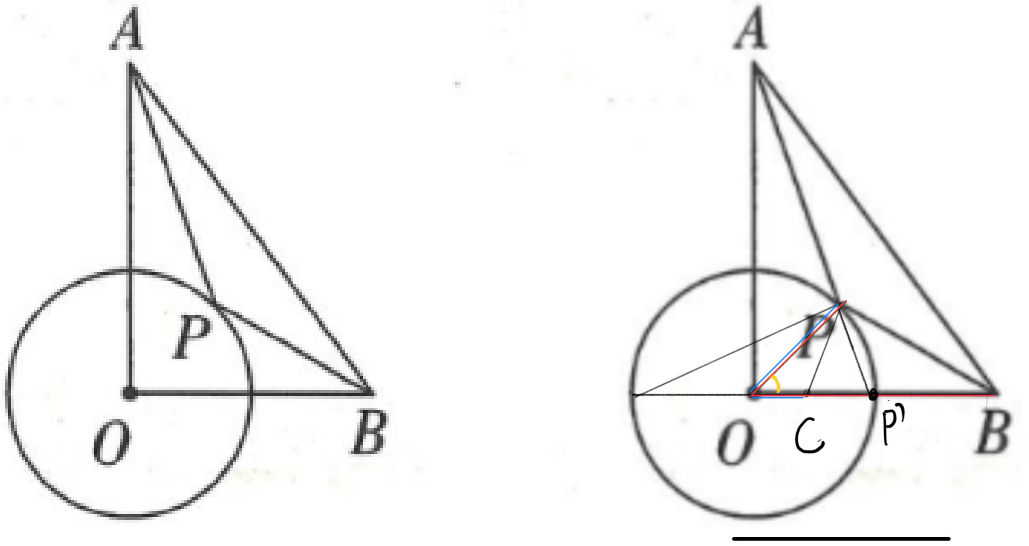


Figure 3: diagram

A city has three logistics hubs: one at point A, one at point B, and one at point O. Your company wants to build a transfer station at point P, 2 miles from the logistics hub at point O. However, the shipping from hub A to station P costs 100 dollars per mile while the shipping from point P to hub B costs 50 dollars per mile. Given that the $\angle AOB$ is a right angle, the distance from hub A to hub O is 4 miles, and the distance from hub B to hub O is 6 miles. How would we determine the optimal location for the transfer station to minimize shipping costs? What is the shipping cost to deliver from hub A to station P and then deliver from station P to hub B? (Hint: We want the minimized value of $100(PA + \frac{PB}{2})$, so we would minimize $PA + \frac{PB}{2}$. How can we convert $\frac{PB}{2}$ into some other segment using Apollonius' circle?)

2.3 Solution!!!

Solution: It is clear that point P can be any point that is 2 units away from point O, which makes all possible location of point P a circle.

Key Observation: There exist a point C on line OB such that $PC = \frac{PB}{2}$.

Reasoning: Although this may seems intuitive to be true for some people, it is actually not easy to prove. Let's first locate point C and review the givens we have. We can first take an extreme case where P is on segment OB, which is also the intersection of circle O and segment OB. To not cause any confusions, we'll name this point P'. We want $PC = \frac{PB}{2}$, so $CP' = \frac{P'B}{2}$. We know that $P'B = BO - P'O$, which yields $P'B = 4 - 2 = 2$. And so point C would be one unit away from point P', such that it is in the interior of the circle.

Now we can review the side lengths, $OC = 1, PO = 2, OB = 4$. We want to create a ratio relationship between PC and PB, and a common way to do it is to prove similarity. Notice that $\angle POB$ is on the opposite of side PC in $\triangle POC$ and is also on the opposite of side PB in $\triangle POB$. Notice that $\frac{OP}{OC} = \frac{OB}{OP} = 2$. Through the SAS similarity Theorem, we can prove that $\triangle POC \sim \triangle BOP$. This implies that $\frac{PB}{PC} = 2$, and we have proven the claim.

Since the side ratio holds true, the circle formed by all possible point P is certainly the Apollonius circle of with fixed point B and C. This means that $PA + \frac{PB}{2} = PA + PC$. This expression is clearly minimized when point P is on line AC. The existence of such point P also have to be taken into consideration. Still, we can make the argument that "all lines that contains points in the interior of a circle intersects the circle", the proof of this will be skipped since this is trivial. If point P is on line AC, $PA + PC = AC$. By Pythagorean

Theorem, we can find the length of AC to be $\sqrt{37}$. The minimized value of $100(PA + \frac{PB}{2})$ will be $100\sqrt{37}$ and we have determined the position of point P.

3 Conclusion

The Apollonius circle provides an elegant geometric configuration that explores ratio-based distance loci. Beyond the theoretical significance, it also offer solutions to pursuit and optimization problems that involves weighted distance. These applications can be extended to robotics path planning, physics, warehouse placements, and more. As a concept that existed for more than 2000 years, it remains significant for its applications and theoretical values.

4 References

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